

NOURISHPAK: AI-BASED CHILD STUNTING & MALNUTRITION RISK PREDICTOR FOR PAKISTAN



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1. Introduction

Pakistan faces one of the most severe child malnutrition crises in the world. Almost 40% of children under five are stunted due to chronic malnutrition, a figure UNICEF's 2023 data puts at 37.6% nationally and as high as 46.6% in Balochistan. Pakistan ranks second-worst in South Asia for child nutrition, with only 48% of infants exclusively breastfed for their first six months.

This is not merely a health issue but a national emergency affecting an estimated 12 million children. Stunting becomes irreversible after age two, and its effects — lower IQ, poorer school performance, and roughly 20% lower lifetime earnings — cost Pakistan an estimated 3% of GDP annually. Compounding the problem, current systems are reactive: health workers identify malnourished children only after visible symptoms appear, by which point the damage is permanent. Rural health workers lack predictive tools, and government nutrition programs distribute resources fairly uniformly despite vastly different malnutrition rates between districts.

The core problem, therefore, is the absence of any low-cost, proactive, and explainable way to identify at-risk children before irreversible harm occurs, and to direct limited government resources to the districts and households that need them most.

We propose NourishPak, a machine learning system that predicts a child's risk of stunting, wasting, underweight, and anemia before damage occurs. Using over 50 tabular features covering the child, mother, household, and environment, the system produces a 0–100 risk score with SHAP-based explanations, works offline on ordinary Android phones, and provides a full Urdu interface — all built on freely available Pakistan Demographic and Health Survey (PDHS) data.

2. Existing System/Related Works

The table below summarises existing systems, apps, and research relevant to child malnutrition tracking and prediction in Pakistan and internationally.

** Year of the system/publication was not specified in the source material and would need to be confirmed before final submission.*

Article/App/System	Year	Features	Strength	Limitations
Mera Doctor (Pakistan)	N/A*	Growth chart plotting; SMS vaccination reminders; Urdu language; free basic tier	Locally available, free, Urdu support	No predictive ML; cannot forecast stunting risk; no geographic mapping; requires internet
Welthungerhilfe's Child Growth Monitor	N/A*	AR/3D smartphone scanning to estimate height and weight; deployed in multiple countries	Camera-based measurement without manual anthropometry	Requires costly AR-capable phone (~\$300+); not available in Pakistan; no Urdu interface; privacy concerns; needs internet
DHIS2 Nutrition Module	N/A*	Used by Pakistan's NIH; tracks vaccination, growth and nutrition enrollment; open-source	Free, government-adopted, multi-language	No predictive analytics; complex interface; no offline app; no explainability or simulation tools
Alive & Thrive Mobile App	N/A*	Nutrition education; breastfeeding/complementary feeding guidance	Deployed across Bangladesh, Vietnam, Ethiopia	Not available for Pakistan/Urdu; no risk prediction or geographic targeting

Article/App/System	Year	Features	Strength	Limitations
Pakistan BISP's Nashonuma Program	N/A*	Conditional cash transfers (PKR 2000/quarter); tracks weight/height at BISP centers; covers 16 districts	Government-backed, direct incentive to families	Detects stunting only after occurrence; limited to 16 districts; no mobile app; cannot predict before age 2

Table 1: Summary of Existing Systems/Work/App

3. Proposed System

NourishPak is a complete AI-powered system designed to predict and prevent child malnutrition by combining three components: a machine-learning Risk Prediction Engine, a Geographic Intelligence Dashboard, and a Health Worker Mobile App. Using tabular data on the child, mother, household, and environment — without requiring any special hardware — it generates a 0–100 risk score with a green/yellow/red indicator, explains the reasoning behind each prediction using SHAP, and provides personalised, affordable nutrition plans. It can also project a child's future growth trajectory to enable early intervention.

3.1. Scope

The system aims to:

- Predict stunting, wasting, underweight, and anemia risk using 50+ tabular features
- Provide SHAP-based explanations for each prediction
- Include a what-if simulator for intervention planning
- Generate district-level risk heatmaps of Pakistan
- Provide an offline-first mobile app for Lady Health Workers
- Operate entirely in the Urdu language
- Use freely available government survey data (PDHS, MICS)
- Require no special hardware — only a basic Android phone

3.2. Proposed System Modules

3.2.1. Risk Prediction Engine

Ingests child, mother, household, and environmental data and runs multiple ML models to output risk scores with SHAP explanations.

- Data ingestion: collect child/mother/household/environmental features
- Model inference: run XGBoost, Random Forest, LightGBM, Logistic Regression, and TabNet models
- Risk scoring: output a 0–100 risk score for stunting, wasting, underweight and anemia
- Explainability: SHAP analysis identifies which features contributed most to each prediction

3.2.2. Geographic Intelligence System

Maps malnutrition risk across Pakistan's districts and supports resource-allocation decisions.

- District-level risk mapping with drill-down from national → province → district → tehsil
- Hotspot detection for clusters of rising malnutrition risk
- Disaster overlay: flood/drought zone data layered onto risk maps
- Resource optimisation: recommend where limited supplies should be directed

3.2.3. Health Worker Mobile App

Gives Lady Health Workers an offline-first, Urdu-language tool for fieldwork.

- Login and case entry: record child data during home visits
- Instant risk prediction: on-device inference via TensorFlow Lite, no internet required
- Personalised nutrition plans: region-specific, affordable food recommendations
- Growth tracking and reminders: vaccination/checkup alerts via SMS and push notification

3.3. Advantage of the Proposed System

- Predicts stunting, wasting, underweight and anemia risk BEFORE irreversible damage occurs (proactive rather than reactive).
- SHAP-based explainability shows health workers exactly why a child is flagged as at-risk.
- Works fully offline on low-end Android phones, unlike existing camera- or internet-dependent tools.
- Complete Urdu-language interface designed for Pakistani Lady Health Workers.
- Built entirely on free, publicly available data (PDHS, MICS, WHO growth standards), keeping the system low-cost.
- District-level geographic risk mapping allows government resources to be targeted where they are most needed.
- "What-if" simulator lets health workers test the effect of interventions before applying them.

3.4. Limitation/Disadvantages

- Training data (PDHS 2017–18) is several years old; malnutrition patterns may have shifted, especially after the 2022 floods.
- Not all rural health workers currently have access to a compatible smartphone.
- Manual data entry by health workers can introduce errors (e.g., mistyped age or weight).
- Model performance may vary across provinces due to differing data distributions (e.g., Balochistan vs. Punjab).

3.5. Application

- Lady Health Workers (LHWs) conducting home visits in rural Pakistan.
- District Health Officers allocating resources to high-risk areas.
- Provincial Health Departments for trend analysis and program evaluation.
- Ministry of National Health Services for national policy and budget planning.
- BISP/Ehsaas program for identifying beneficiaries of conditional cash transfers.
- NGOs (UNICEF, WHO, Save the Children) for targeted nutrition interventions.
- Researchers working in public health and machine learning for global health.

3.6. SDG Goals

The proposed system was not mapped to specific SDGs in the original report; based on its objectives, the most relevant goals are:

- SDG 2 — Zero Hunger: directly targets malnutrition, stunting and wasting in children.
- SDG 3 — Good Health and Well-being: enables early, preventive healthcare intervention for children under five.
- SDG 1 — No Poverty: supports targeted cash-transfer and resource-allocation programs (e.g., BISP/Ehsaas) toward the most vulnerable households.
- SDG 10 — Reduced Inequalities: geographic targeting directs limited government resources to the highest-need districts rather than distributing them uniformly.

4. Hardware/Software Requirements

Software

Category	Tools
ML Development	Python, Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, SHAP, Lifelines (Survival Analysis), TensorFlow lite
Backend	FastAPI, PostgreSQL, Firebase
Frontend	React.js, Plotly, Leaflet/Mapbox, GeoPandas
Mobile App	Flutter (cross-platform), TensorFlow Lite (on-device ML)
Reporting	WeasyPrint (PDF generation)
Data Sources	PDHS dataset, MICS dataset, WHO Growth Standards, PBS shapefiles, NDMA disaster data

Hardware

Hardware	Specifications
Development PC	Core i7 8th Gen+, 16GB RAM, SSD (for model training)
Mobile Device (Testing)	Android 9.0+ (any budget phone, e.g., Infinix, Oppo, Tecno)
Deployment Server	Cloud VM (AWS/Azure) or local server for dashboard hosting

5. References

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